

ARIXTRA 5MG/0.4ML

ARIXTRA 7.5MG/0.6ML

Fondaparinux sodium

QUALITATIVE AND QUANTITATIVE COMPOSITION

Sterile, preservative-free, clear and colourless to slightly yellow, injectable solution, in a single dose pre-filled syringe.

Each syringe contains 5.0 mg of fondaparinux sodium in 0.4 ml solution for injection.

Each syringe contains 7.5 mg of fondaparinux sodium in 0.6 ml solution for injection.

PHARMACEUTICAL FORM

Injectable solution for subcutaneous use.

CLINICAL PARTICULARS

Therapeutic Indications

Treatment of acute Deep Vein Thrombosis (DVT).

Treatment of acute Pulmonary Embolism (PE).

Posology and Method of Administration

Method of administration

Subcutaneous administration

The sites of administration should alternate between the left and the right anterolateral and left and right posterolateral abdominal wall. To avoid the loss of medicinal product when using the pre-filled syringe do not expel the air bubble from the syringe before the injection. The whole length of the needle should be

inserted perpendicularly into a skin fold held between the thumb and the forefinger. The skin fold should be held throughout the injection.

ARIXTRA is intended for use under a physician's guidance. Patients may self-inject only if their physician determines that it is appropriate, and with medical follow-up as necessary. Proper training in subcutaneous injection technique should be provided. Instruction for self-administration is included in the package leaflet (see *Instructions for Use/Handling*).

- **Adults**

TREATMENT OF DVT AND PE

The recommended dose of *ARIXTRA* to be administered by subcutaneous injection once daily is:

- 5 mg for body weight less than 50 kg;
- 7.5 mg for body weight 50 to 100 kg;
- 10 mg for body weight greater than 100 kg.

Treatment should be continued for at least 5 days and until adequate oral anticoagulation is established (International Normalised Ratio 2 to 3).

Concomitant treatment with vitamin K antagonists should be initiated as soon as possible, usually within 72 hours. The usual duration of *ARIXTRA* treatment is 5 to 9 days (see *Clinical Studies*).

Special Populations

- **Children**

The safety and efficacy of *ARIXTRA* in patients under the age of 17 has not been established (see *Clinical Studies*).

- **Elderly (from 75 years)**

ARIXTRA should be used with caution in elderly patients as renal function decreases with age (see *Renal impairment, Special Warnings and Precautions for Use*).

- **Patients with body weight less than 50 kg**

Patients with body weight below 50 kg are at increased risk of bleeding (see *Special Warnings and Precautions for Use*).

- **Renal impairment**

ARIXTRA should not be used in patients with a creatinine clearance less than 30 ml/min (see *Special Warnings and Precautions for Use and Pharmacokinetic Properties*). No dosage reduction is required for patients with a creatinine clearance greater than or equal to 30ml/min.

- **Hepatic impairment**

No dosing adjustment of *ARIXTRA* is necessary in patients with mild to moderate hepatic impairment (see *Pharmacokinetic Properties*). In patients with severe hepatic impairment, *ARIXTRA* should be used with caution (see *Special Warnings and Precautions for Use*).

Contraindications

- known hypersensitivity to *ARIXTRA* or any of the excipients.
- active clinically significant bleeding.
- acute bacterial endocarditis.
- severe renal impairment (creatinine clearance less than 30 ml/min).

Special Warnings and Precautions for Use

Route of administration - *ARIXTRA* must not be administered intramuscularly (see *Posology and Method of Administration*).

Haemorrhage - *ARIXTRA*, like other anticoagulants, must be used with caution in conditions with an increased risk of haemorrhage, (such as congenital or acquired bleeding disorders, active ulcerative gastrointestinal disease, recent intracranial haemorrhage, shortly after brain, spinal or ophthalmic surgery).

- ***Treatment of VTE***

Other medical products enhancing the risk of haemorrhage, with the exception of vitamin K antagonists used concomitantly for treatment of VTE, should not be administered with *ARIXTRA*. If co-administration is essential, close monitoring is recommended (see *Interaction with Other Medicinal Products and Other Forms of Interaction*).

Spinal/epidural anaesthesia/spinal puncture - Epidural or spinal haematomas that may result in long-term or permanent paralysis can occur with the use of anticoagulants and spinal/epidural anaesthesia or spinal puncture. The risk of these rare events may be higher with post-operative use of indwelling epidural catheters or the concomitant use of other medicinal products affecting haemostasis.

Elderly patients - The elderly population is at increased risk of bleeding. As renal function generally decreases with age, elderly patients may show reduced elimination and increased exposure of *ARIXTRA*. *ARIXTRA* should be used with caution in elderly patients (see *Posology and Method of Administration*).

Low body weight - Patients with body weight less than 50 kg are at increased risk of bleeding. Elimination of *ARIXTRA* decreases with weight decrease. *ARIXTRA* should be used with caution in these patients (see *Posology and Method of Administration*).

Renal impairment - The plasma clearance of fondaparinux decreases with the severity of renal impairment and is associated with an increased risk of haemorrhage (see *Pharmacokinetic Properties*).

Patients with renal impairment, particularly those with a creatinine clearance of less than 30ml/min are at increased risk of both major bleeding episodes and VTE.

Severe hepatic impairment - In patients with an elevation in prothrombin time, the use of *ARIXTRA* should be considered with caution, because of an increased risk of bleeding due to a possible deficiency of coagulation factors in patients with severe hepatic impairment (see *Posology and Method of Administration*).

Heparin Induced Thrombocytopenia - *ARIXTRA* does not bind to platelet factor 4 and does not cross-react with sera from patients with Heparin Induced Thrombocytopenia (HIT)-type II. It should be used with caution in patients with a history of HIT. The efficacy and safety of *ARIXTRA* have not been formally studied in HIT-type II. Rare spontaneous reports of HITS in patients treated with *ARIXTRA* have been received. To date a causal association between treatment with *ARIXTRA* and the occurrence of HIT has not been established.

Latex Allergy – The needle shield of the pre-filled syringe contains dry natural latex rubber that has the potential to cause allergic reactions in latex sensitive individuals.

Interaction with Other Medicinal Products and Other Forms of Interaction

Fondaparinux does not markedly inhibit CYP450s (CYP1A2, CYP2A6, CYP2C9, CYP2C19, CYP2D6, CYP2E1 or CYP3A4) in vitro. Thus, *ARIXTRA* is not expected to interact with other medicinal products in vivo by inhibition of CYP-mediated metabolism.

Since fondaparinux does not bind significantly to plasma proteins other than ATIII, no interaction with other medicinal products by protein binding displacement is expected.

In clinical studies performed with fondaparinux, the concomitant use of warfarin (oral anticoagulant), acetylsalicylic acid (platelet inhibitor), piroxicam (non-steroidal anti-inflammatory), and digoxin (cardiac glycoside) did not significantly affect the pharmacokinetics or pharmacodynamics of fondaparinux. In addition, fondaparinux neither influenced the INR activity of warfarin, nor the bleeding time under acetylsalicylic acid or piroxicam treatment, nor the pharmacokinetics or pharmacodynamics of digoxin at steady state.

Pregnancy and Lactation

Pregnancy

There are limited clinical data available on exposed pregnancies. *ARIXTRA* should not be prescribed to pregnant women unless the benefit outweighs the risk (see *Non-Clinical Information*).

Lactation

Fondaparinux is excreted in rat milk but it is not known whether fondaparinux is excreted in human milk. Breast-feeding is not recommended during treatment with *ARIXTRA*.

Effects on Ability to Drive and Use Machines

No studies on the effect on the ability to drive and to use machines have been performed.

Undesirable Effects

The safety of fondaparinux has been evaluated in:

- 3,595 patients undergoing major orthopaedic surgery of the lower limbs treated up to 9 days (Arixtra 1.5 mg/0.3 ml and Arixtra 2.5 mg/0.5 ml)
- 327 patients undergoing hip fracture surgery treated for 3 weeks following an initial prophylaxis of 1 week (Arixtra 1.5 mg/0.3 ml and Arixtra 2.5 mg/0.5 ml)
- 1,407 patients undergoing abdominal surgery treated up to 9 days (Arixtra 1.5 mg/0.3 ml and Arixtra 2.5 mg/0.5 ml)
- 425 medical patients who are at risk for thromboembolic complications treated up to 14 days (Arixtra 1.5 mg/0.3 ml and Arixtra 2.5 mg/0.5 ml)
- 10,057 patients undergoing treatment of UA or NSTEMI ACS (Arixtra 2.5 mg/0.5 ml)
- 6,036 patients undergoing treatment of STEMI ACS (Arixtra 2.5 mg/0.5 ml)
- 2,517 patients treated for Venous Thrombo-Embolicism and treated with fondaparinux for an average of 7 days (Arixtra 5 mg/0.4 ml, Arixtra 7.5 mg/0.6 ml and Arixtra 10 mg/0.8 ml).

These adverse reactions should be interpreted within the surgical or medical context of the indications. The adverse event profile reported in the ACS program is consistent with the adverse drug reactions identified for VTE prophylaxis.

Adverse reactions are listed below by system organ class and frequency. Frequencies are defined as: very common ($\geq 1/10$), common ($\geq 1/100$, $<1/10$), uncommon ($\geq 1/1,000$, $<1/100$), rare ($\geq 1/10,000$, $<1/1,000$), very rare ($<1/10,000$).

System organ class MedDRA	common (≥ 1/100, <1/10)	uncommon (≥ 1/1,000, <1/100)	rare (≥ 1/10,000, <1/1,000)
Infections and infestations			post-operative wound infections
Blood and lymphatic system disorders	anaemia, post-operative haemorrhage, utero-vaginal haemorrhage*, haemoptysis, haematuria, haematoma, gingival bleeding, purpura, epistaxis, gastrointestinal bleeding, hemarthrosis*, ocular bleeding*, bruise*	thrombocytopenia, thrombocythaemia, platelet abnormal, coagulation disorder	retroperitoneal bleeding*, hepatic, intracranial/ intracerebral bleeding*
Immune system disorders			allergic reaction (including very rare reports of angioedema, anaphylactoid/ anaphylactic reaction)
Metabolism and nutrition disorders			hypokalaemia, non-protein-nitrogen (Npn) increased ^{1*}
Nervous system Disorders		headache	anxiety, confusion, dizziness, somnolence, vertigo
Vascular disorders			hypotension
Respiratory, thoracic and mediastinal disorders		dyspnoea	coughing
Gastrointestinal Disorders		nausea, vomiting	abdominal pain, dyspepsia, gastritis, constipation, diarrhoea
Hepatobiliary disorders		abnormal liver function tests, hepatic enzymes increased	bilirubinaemia
Skin and subcutaneous		rash erythematous, pruritus	

tissue disorders			
General disorders and administration site conditions		oedema, oedema peripheral, pain, fever, chest pain, wound secretion	reaction at injection site, leg pain, fatigue, flushing, syncope, hot flushes, oedema genital

⁽¹⁾ Npn stands for non-protein-nitrogen such as urea, uric acid, amino acid, etc.

*ADRs occurred at higher doses 5 mg/0.4 ml, 7.5 mg/0.6 ml and 10 mg/0.8 ml.

Overdose

Symptoms and Signs

ARIXTRA doses above the recommended regimen may lead to an increased risk of bleeding.

Treatment

Overdose associated with bleeding complications should lead to treatment discontinuation and search for the primary cause. Initiation of appropriate therapy which may include surgical haemostasis, blood replacements, fresh plasma transfusion, plasmapheresis should be considered.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic Properties

Pharmacotherapeutic group: antithrombotic agents.

ATC Code

B01AX05

Mechanism of Action

Fondaparinux is a synthetic and selective inhibitor of activated Factor X (Xa). The antithrombotic activity of fondaparinux is the result of antithrombin III (ATIII) mediated selective inhibition of Factor Xa. By binding selectively to ATIII, fondaparinux potentiates (about 300 times) the innate neutralization of Factor Xa by ATIII. Neutralisation of Factor Xa interrupts the blood coagulation cascade and inhibits both thrombin formation and thrombus development.

Fondaparinux does not inactivate thrombin (activated Factor II) and has no known effect on platelet function.

Pharmacodynamic Effects

At the 2.5 mg dose, fondaparinux does not have a clinically relevant effect on routine coagulation tests, such as activated partial thromboplastin time (aPTT),

activated clotting time (ACT) or prothrombin time (PT)/International Normalised Ratio (INR) tests in plasma, nor bleeding time or fibrinolytic activity. However, rare spontaneous reports of elevated aPTT have been received at the 2.5mg dose.

Fondaparinux does not cross-react with sera from patients with Heparin Induced Thrombocytopenia (HIT) type II.

Anti-Xa activity

The pharmacodynamics/pharmacokinetics of fondaparinux are derived from fondaparinux plasma concentrations quantified via anti factor Xa activity. Only fondaparinux can be used to calibrate the anti-Xa assay. The international standards of heparin or low molecular weight heparin (LMWH) are not appropriate for this use. As a result, the concentration of fondaparinux is expressed as milligrams of the fondaparinux calibrator/litre.

Pharmacokinetic Properties

Absorption

After subcutaneous dosing, fondaparinux is completely and rapidly absorbed (absolute bioavailability 100%). Following a single subcutaneous injection of *ARIXTRA* 2.5 mg to young healthy subjects, peak plasma concentration, mean C_{max} of 0.34 mg/L, is reached in approximately 2 hours. Plasma concentrations of half the mean C_{max} values are reached 25 min post-dosing.

In elderly healthy subjects, pharmacokinetics of fondaparinux are linear in the range of 2 to 8 mg by subcutaneous route. Following once daily dosing, steady state of plasma levels is obtained after 3 to 4 days with a 1.3-fold increase in C_{max} and AUC. Following a single i.v. bolus administration to healthy elderly subjects, the pharmacokinetics of fondaparinux are linear over the therapeutic range.

In patients undergoing hip replacement surgery receiving *ARIXTRA* 2.5 mg once daily, the peak steady-state plasma concentration is, on average, 0.39 to 0.50 mg/L and is reached approximately 3 hours post-dose. In these patients, the minimum steady-state plasma concentration is 0.14 to 0.19 mg/L.

In patients with symptomatic deep vein thrombosis and pulmonary embolism undergoing treatment with *ARIXTRA* 5 mg (body weight less than 50 kg), 7.5 mg (body weight 50 to 100 kg) and 10 mg (body weight greater than 100 kg) once daily, the body-weight-adjusted doses provide similar mean steady-state peaks and minimum plasma concentrations across all body weight categories. The mean peak steady-state plasma concentration is in the range of 1.20 to 1.26 mg/L. In these patients, the mean minimum steady-state plasma concentration is in the range of 0.46 to 0.62 mg/L.

Distribution

In healthy adults, subcutaneously administered fondaparinux distributes mainly in blood and only to a minor extent in extravascular fluid, as demonstrated by steady

state and non-steady state apparent volume of distribution of 7 to 11 L. *In vitro*, fondaparinux is highly (at least 94%) and specifically bound to antithrombin III (ATIII) and does not bind significantly to other plasma proteins, including platelet Factor 4 (PF4) or red blood cells.

Metabolism

In vivo metabolism of fondaparinux has not been investigated since the majority of the administered dose is eliminated unchanged in urine in individuals with normal kidney function.

Elimination

Fondaparinux is eliminated in urine mainly as unchanged drug. In healthy individuals, 64 to 77% of a single dose is eliminated in urine in 72 hours. The elimination half-life is about 17 hours in healthy young subjects and about 21 hours in healthy elderly subjects. In patients with normal renal function, the mean fondaparinux clearance is 7.82mL/min.

Special Patient Populations

- **Renal impairment**

Fondaparinux elimination is prolonged in patients with renal impairment since the major route of elimination is urinary excretion of unchanged drug. In patients undergoing prophylaxis following elective hip surgery or hip fracture surgery, the total clearance of fondaparinux is approximately 25% lower in patients with mild renal impairment (creatinine clearance 50 to 80 ml/min), approximately 40% lower in patients with moderate renal impairment (creatinine clearance 30 to 50 ml/min) and approximately 55% lower in patients with severe renal impairment (less than 30 ml/min), compared to patients with normal renal function. The associated terminal half-life values were 29 hours in moderate and 72 hours in patients with severe renal impairment. A similar relationship between fondaparinux clearance and extent of renal impairment was observed in DVT treatment patients. Due to the limited clinical data available, *ARIXTRA* should not be used in patients with severe renal impairment (see *Special Warnings and Precautions for Use*).

- **Hepatic impairment**

Unbound concentrations of fondaparinux are expected to be unchanged in patients with mild to moderate hepatic impairment, and therefore, no dose adjustment is necessary based on pharmacokinetics. Following a single, subcutaneous dose of fondaparinux in subjects with moderate hepatic impairment (Child-Pugh Category B), C_{max} and AUC were decreased by 22% and 39%, respectively, as compared to subjects with normal liver function. The lower plasma concentrations of fondaparinux were attributed to reduced binding to ATIII secondary to the lower ATIII plasma concentrations in subjects with hepatic impairment thereby resulting in increased renal clearance of fondaparinux.

The pharmacokinetics of *ARIXTRA* has not been studied in patients with severe hepatic impairment (see *Posology and Method of Administration, Special*

Warnings and Precautions for Use).

- **Children**

The use of *ARIXTRA* has not been investigated in children under the age of 17 years.

- **Elderly**

Fondaparinux elimination is prolonged in patients over 75 years old. In studies evaluating *ARIXTRA* 2.5 mg prophylaxis in hip fracture surgery or elective hip surgery, the total clearance of fondaparinux was approximately 25% lower in patients over 75 years old as compared to patients less than 65 years old. A similar relationship between fondaparinux clearance and age was observed in DVT treatment patients.

- **Gender**

No gender differences were observed after adjustment for body weight.

- **Race**

Pharmacokinetic differences due to race have not been studied prospectively. However, studies performed in Asian (Japanese) healthy subjects did not reveal a different pharmacokinetic profile compared to Caucasian healthy subjects. Similarly, based on the results of population pharmacokinetic analysis conducted in patients undergoing orthopaedic surgery, no plasma clearance differences were observed between black and Caucasian patients.

- **Body weight**

In patients weighing less than 50 kg the total clearance of fondaparinux sodium is decreased by approximately 30% (*see Special Warnings and Precautions for Use*).

Clinical Studies

Treatment of deep vein thrombosis (DVT) and pulmonary embolism (PE)

- **DVT**

In patients with a confirmed diagnosis of acute symptomatic DVT, *ARIXTRA* 5 mg (body weight less than 50 kg), 7.5 mg (body weight 50 kg to 100 kg) or 10 mg (body weight greater than 100 kg) once daily, was compared to enoxaparin 1 mg/kg subcutaneously twice daily. Patients were treated for at least 5 days in conjunction with a vitamin K antagonist which was continued for 90 ± 7 days, with regular dose adjustments to achieve an INR of 2 to 3.

ARIXTRA was demonstrated to be non-inferior to enoxaparin (VTE rates 3.9% and 4.1% at Day 97, respectively). Major bleeding during the initial treatment period was observed in 1.1% of *ARIXTRA* patients, compared to 1.2% with enoxaparin.

- **PE**

In patients with a confirmed diagnosis of acute symptomatic PE, *ARIXTRA* 5 mg (body weight less than 50 kg), 7.5 mg (body weight 50 kg to 100 kg) or 10 mg (body weight greater than 100 kg) once daily, was compared to unfractionated heparin (UFH) i.v. bolus (5000 IU), followed by a continuous iv infusion adjusted to maintain 1.5 to 2.5 times aPTT control value. Patients were treated for at least 5 days in conjunction with a Vitamin K antagonist which was continued for 90 ± 7 days, with regular dose adjustments to achieve an INR of 2 to 3.

ARIXTRA was demonstrated to be non-inferior to UFH (VTE rates 3.8% and 5.0% at Day 97, respectively). Major bleeding during the initial treatment period was observed in 1.3% of *ARIXTRA* patients, compared to 1.1% with UFH.

Pre-clinical Safety Data

No long-term studies in animals have been performed to evaluate the carcinogenic potential of fondaparinux sodium.

Fondaparinux sodium was not genotoxic in the Ames test, the mouse lymphoma cell (L5178Y/TK+/-) forward mutation test, the human lymphocyte chromosome aberration test, the rat hepatocyte unscheduled DNA synthesis (UDS) test, or the rat micronucleus test.

Reproduction studies have been performed in rats and rabbits at subcutaneous doses up to 10 mg/kg/day (approximately 5 and 12 times human exposure at a dose of 2.5mg, or 2 and 4 times human exposure at a dose of 7.5mg, based on AUC) and have revealed no evidence of impaired fertility or harm to the foetus due to fondaparinux sodium. Because animal reproduction studies are not always predictive of human response, *ARIXTRA* should not be prescribed to pregnant women unless the risk of VTE outweighs the potential risk to the foetus.

PHARMACEUTICAL PARTICULARS

List of Excipients

Sodium chloride

Water for injection

Hydrochloric acid or sodium hydroxide for pH adjustment as necessary.

Incompatibilities

In the absence of compatibility studies, *ARIXTRA* must not be mixed with other medicinal products.

Shelf Life

The expiry date is indicated on the packaging.

Special Precautions for Storage

Store below 30°C. Do not freeze.

Nature and Contents of Container

ARIXTRA pre-filled single-use syringes are made of Type I glass barrel (1 ml) affixed with a 27 gauge x 12.7 mm needle and stoppered with a bromobutyl or chlorobutyl elastomer plunger stopper.

ARIXTRA 5.0mg/0.4ml is available in pack sizes of 2 and 10 pre-filled syringes with an orange plunger and an automatic safety system.

ARIXTRA 7.5 mg/0.6 ml is available in pack sizes of 2 and 10 pre-filled syringes with a magenta plunger and an automatic safety system.

Not all presentations are available in every country.

Instructions for Use/Handling

Parenteral solutions should be inspected visually for particulate matter and discoloration prior to administration.

ARIXTRA is administered by subcutaneous injection or intravenous injection. It must not be administered by intramuscular injection. The subcutaneous injection is administered in the same way as with a standard syringe.

The *ARIXTRA* pre-filled syringe has been designed with an automatic needle protection system to prevent needle stick injuries following injection.

Instruction for self-administration by subcutaneous injection is included in the package leaflet.

Any unused product or waste material should be disposed of in accordance with local requirements.

Step-by-step instructions

Parts of the syringes:

- ① Needle shield
- ② Plunger
- ③ Finger-grip
- ④ Security sleeve



Instructions for use

1. Wash your hands thoroughly with soap and water and dry them with a towel.

2. Remove the syringe from the carton and check that:

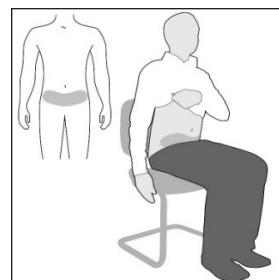
- the expiry date has not passed
- the solution is clear and colourless to slightly yellow and doesn't contain particles
- the syringe has not been opened or damaged.

3. Sit or lie down in a comfortable position.

Choose a place in the lower abdominal (tummy) area, at least 5 cm below your belly button (picture **A**).

Alternate the left and right side of the lower abdominal area at each injection. This will help to reduce the discomfort at the injection site.

If injecting in the lower abdominal area is not possible, ask your nurse or doctor for advice.

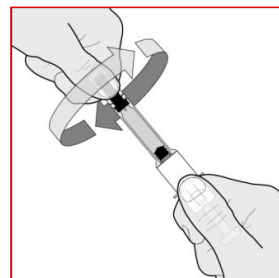


Picture A

4. Clean the injection area with an alcohol wipe.

5. Remove the needle shield, by first twisting it (picture **B1**) and then pulling it in a straight line away from the body of the syringe (picture **B2**).

Discard the needle shield.

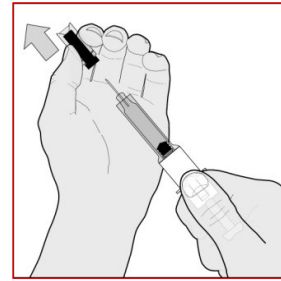


Picture B1

Important note:

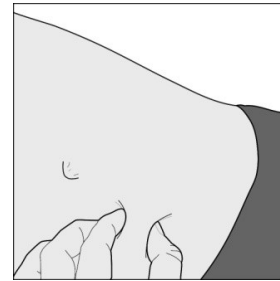
- Don't touch the needle or allow it to touch any surface before the injection

It is normal to see a small air bubble in this syringe. Don't try to remove this air bubble before making the injection - you may lose some of the medicine if you do.



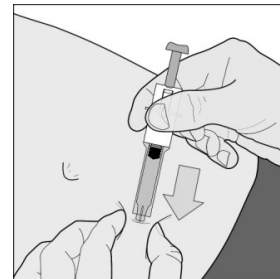
Picture B2

6. Gently pinch the skin that has been cleaned to make a fold. Hold the fold between the thumb and the forefinger during the entire injection (picture C).



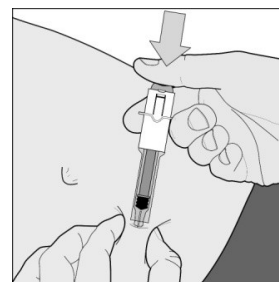
Picture C

7. Hold the syringe firmly by the finger grip. Insert the full length of the needle at right angles into the skin fold (picture D).



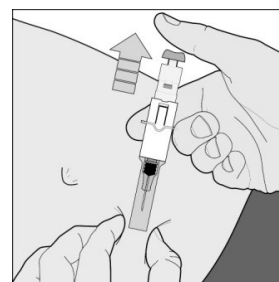
Picture D

8. Inject ALL of the contents of the syringe by pressing down on the plunger as far as it goes (picture E).



Picture E

9. Release the plunger and the needle will automatically withdraw from the skin and go back into the security sleeve where it will be locked permanently (picture F).



Picture F

Do not dispose of the used syringe in household waste.
Dispose of it as your doctor or pharmacist has instructed.

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MANUFACTURER

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